EEN1095 Project Meeting Minutes

Student Name	Darshan Mahadeva Naika	**Student ID**	A00090581
Project Title	Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels	**Supervisor**	Martin Collier
Implementation Period	Week 1 - Week 2	**Version Date**	22/05/2026

Guidance: Use this template whenever a project meeting takes place. Please note that no formally scheduled meetings are expected during the independent implementation period. However, any meetings that occur should be recorded appropriately.

Date	22/05/2026	**Time**	12:00 pm to 12:20 pm
Location	Zoom (Online)	**Attendees**	Darshan Mahadeva Naika, Martin Collier

1. Agenda (What was planned)

- E.g. Review of progress
- Technical guidance on filter implementation — lossless discrete integrator and s-plane transformation.

2. Key Discussion Points

- **Topic A:**
- **Topic B:**
- **Topic C:**

3. Decisions Made

Start implementation with Direct-Form filter as the baseline structure. Derive the lossless discrete integrator transformation mathematically and compute coefficient values before coding. Use the experimental order: low precision first (float16, float32) then high precision (float64, mpmath). Investigate Python arbitrary precision libraries and evaluate whether they give measurably better results. Define sensitivity formally before measuring it in the framework.

4. Action Items (To be completed by next meeting)

Task	Deadline	Status
1. Study the lossless discrete integrator transformation from s-plane to z-plane. Derive the coefficient values mathematically. Validate using a simulator by generating the impulse response and feeding it into the difference equation to obtain the z-transform.	By next meeting	**In Progress**
2. Implement Direct-Form IIR filter as the baseline structure. Run	By next meeting	**To Do**

Task	Deadline	Status
experiments starting with float16 (low precision) first to observe instability, then float32, float64, and mpmath. Document results at each precision level.		
3. Investigate Python arbitrary precision libraries. Assess whether mpmath and similar libraries provide measurably better filter performance results compared to standard float64. Define filter sensitivity formally and implement its measurement in the framework.	By next meeting	**To Do**

5. **Next Meeting**
   - Date: To be scheduled